

RBSE Class 11 Chemistry - Chapter 1

Class 11 Chemistry RBSE — Chapter 1 Some Basic Concepts of Chemistry

1. Chemistry

Chemistry is the branch of science that studies matter, its composition, properties and changes.

2. Matter

Anything that has mass and occupies space is called matter.

3. Classification of Matter

- Solid
- Liquid
- Gas

4. Laws of Chemical Combination

- Law of Conservation of Mass
- Law of Constant Proportion
- Law of Multiple Proportion

5. Dalton's Atomic Theory

Matter is made of atoms. Atoms combine in simple ratios.

6. Atom and Molecule

Atom is the smallest particle of element.

Molecule is combination of atoms.

7. Atomic Mass

Mass of atom compared to carbon atom.

8. Molecular Mass

Example:

$$\text{H}_2\text{O} = (2 \times 1) + 16 = 18 \text{ u}$$

9. Mole Concept

$$1 \text{ mole} = 6.022 \times 10^{23} \text{ particles}$$

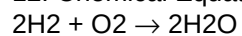
10. Formula for Mole

$$\text{Mole} = \text{Given Mass} / \text{Molar Mass}$$

11. Percentage Composition

$$\% \text{ of element} = (\text{Mass of element} / \text{Molecular mass}) \times 100$$

12. Chemical Equation



13. Stoichiometry

Calculations based on chemical equations.

Importance:

This chapter is the base of chemistry and very important for RBSE board exams.